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Cybercrime in the context of the digital age: analysis of threats, legal challenges and strategies

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Abstract The current period of social evolution is commonly referred to as the information stage of society's development or the digital age. The introduction of the most advanced tools significantly speeds up the process of obtaining, processing and analyzing data. Wide and prompt access to information resources increases the efficiency of their use, which becomes an integral part of managing all institutions and processes. Various forms of crime are emerging globally as science and information technology evolve. The emergence and development of cybercrime is taking place, in particular, due to the globalization of the information environment and the rapid development of computer systems around the world. Currently, information technologies accompany people in various spheres of their activities, and accordingly, their use in committing criminal offences is increasing. In connection with the abovementioned, the issues of legal support for information security occupy a specific place in the public relations spectrum. The topic of cybersecurity in the era of information technology is increasingly spreading in the world community nowadays. After all, because of technological development, the number of cybercrimes that disrupt the activities of states, their institutions and a large number of private companies is rapidly increasing, with institutions and organizations incurring significant costs and disrupting their normal operation. The purpose of the academic paper is to define the concept and to review the classifications of cybercrime, taking into account the features of its commission and counteraction to this type of abuse in the context of digital society.

Keywords: cybercrime, cyberspace, counteraction to cybercrime, cyberattack, company's cybersecurity, European cybersecurity policy.

1. Introduction

A high level of information security is critically important for the growth of modern society. The proliferation of new information technologies based on the widespread use of computer systems and means of communication, optimization and automation of processes has led to the search for new ways to securely use information and communication systems in various spheres of society. Furthermore, the adoption of a well-balanced governmental cybercrime strategy and effective legal regulation of the use of information resources are crucial components of guaranteeing information security.

The concepts, features, and relevance of the fight against cybercrime are considered in the theoretical part of the present research, and their types and key areas of counteraction by public and private structures are substantiated.

The practical part of the academic paper includes an overview of the number of cybercrimes committed against information and computer systems of companies around the world. The research describes the time stages of developing legal regulation of the counteraction to cybercrime in the world and highlights the main features of legal regulation in the fight against cybercrime in the European Union.

Based on the results of the research, conclusions were drawn on the issues raised. In particular, the most significant aspects that are prerequisites for the threats of spreading cybercrime have been highlighted. Along with this, the conditions for effective counteraction to cybercrime have been outlined and the prerequisites and factors that pose threats to cyberspace security in most countries of the world have been identified.

The purpose of the practical part of the research is to identify areas of geographical spread of cybercrime, taking into account worldwide trends in globalization and digitalization of society. The objectives of studying the practical aspects of the research topic include as follows: establishing the vectors of cybercrime development in various nations, exploring academic

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opinions on the legal regulation of the process of combating crimes in cyberspace and current trends in the regulation outlined, and identifying the key elements of combating cybercrime in the territory of the European Union.

2. Literature review

Cyber security crimes are currently among the most dangerous and numerous since they are new compared to other types of crime, and the pace of development of systems to combat this type of illegal acts lags far behind the speed and number of their occurrence and spread (Danich, 2019).

Cybercrime is a crime committed with the help of or through computer systems, violating the right of information environment subjects to protect information, resulting in negative consequences of impact on information or the process of its maintenance (Petrenko et al., 2022; Bondarenko, et al., 2022). The above-mentioned type of crime also involves other types of socially dangerous actions. These may be related to non-compliance with the legislation on information ownership or the established procedure for using information technologies, the rights of the owner or user of information data to timely receive or disseminate accurate and complete information (Chatterjee and Thekdi, 2020; Danich, 2019; Lytvyn, 2022).

The term "cybercrime" refers to crimes committed using an electronic network or against such a network, systems or networks, or their owners. This area of illegal acts involves three categories of unlawful acts:

1) traditional forms of crime (various types of fraud or forgery of information in the electronic network;
2) posting of illegal content;
3) specific types of criminal or illegal acts committed in an electronic computer network (various types of attacks against an information system, hacking, etc.) (Yacyk et al., 2019; Qian et al., 2023).

The current increase in the quantitative and qualitative composition of crimes and their types in cyberspace can be considered a consequence of improved technical and software tools available to criminals, as well as the rapid development of illegal markets for the sale of tools for committing illegal actions in cyberspace. Scientists note that cybercrime, compared to traditional crime, has several distinctive features, including as follows:

- the implementation of cybercrime in terms of location, compared to traditional types of crime, can occur within different jurisdictions (for instance, in cases where the offender activates a cyberattack from an Internet cafe in one country, a botnet – in another country, and the affected resource – in the third country);
- most facts of evidence of cybercrime are in electronic form ("digital" evidence). As opposed to conventional ones, they can be quickly destroyed or changed. Special devices are required to collect, store, and analyze them;
- due to the specifics of cyberspace, the victim does not always realize that a cybercrime has been committed (Heartfeld et al., 2021; Nikulesko, 2019).

Cybercrimes include infringement of copyright and related rights, various types of fraud, illegal activities that may be related to the transfer of funds, payment card transactions and unauthorized access to a bank account, tax evasion, and the use of information that is a commercial or banking secret (Habib et al., 2020; Iaiani et al., 2022; Novak, 2022).

The most widespread types of this type of crime are as follows:

- Carding – illegal operations with payment card data.
- Phishing is a type of fraud in which a customer of a payment system is sent a request for his account.
- Vishing is a form of cybercrime in which confidential information is stolen during a call.
- Online (Internet) fraud is a type of illegal activity that uses fake online auctions, online stores, and websites.
- Piracy is the unlawful conduct of transactions on the Internet about intellectual property.
- Cardsharing is the illegal access to satellite and cable television.
- Social engineering tools are means of unlawful management of human resources via the Internet.
- Illegal software means the development or distribution of viruses and malware.
- Illegal content is the dissemination of information that promotes extremism, terrorism, etc.
- Refiling is an illegal interference with telephone traffic (Danich, 2019).

The legal regulation of the fight against cybercrime in most EU countries is characterized by the following features:

- availability of both national and international legislation to combat cybercrime;
- activities to combat cybercrime are carried out simultaneously by national and international organizations, which include specialists from participating countries;
- theoretical issues, such as expert assessment of cybercrime, development of advanced methods of prevention and investigation, etc. play an important role in the fight against cybercrime;
- conducting an active exchange of information (Guo et al., 2020).

In the field of combating cybercrime in recent years, the following tendencies in developing the global theoretical, research and legislative framework are observed:

- acquisition of a new, broader meaning of the term "cybercrime";
- increasing responsibility for committing this type of crime;
- harmonization of approaches to the definition of concepts related to cybercrime in all legal acts regulating this issue.

It should be noted that the legislation of highly developed countries has made a significant step toward the development and implementation of the regulatory framework governing relations arising from the use of information technology in recent years (Bayev et al., 2022; Borodina et al., 2022). At the same time, the analytical and legal framework for regulating cybersecurity is constantly being supplemented and improved in the states, taking into account the rapidly changing technological and social conditions for the provision of electronic services. However, in the vast majority of developing countries, the vector of regulation in this area has not been determined yet (Akimova et al., 2022; Akimov et al., 2021; Zayed, 2022).

There is no doubt that the legal regulation of relations arising from the use of Internet technologies, in particular, and the overall strategic management of the processes of preventing, combating and countering the consequences of cybercrime are still controversial areas and remain insufficiently studied. Consequently, they require scientific study and increased attention at both the national and international levels (Danich, 2019).

3. Methods

A practical study of current trends in cybercrime development was conducted using the statistical data analysis, comparison and generalization of the information obtained. The information base of the paper is composed of the statistical data provided in the International Journal of Scientific Research and Management, and the results of the work of scientists and lawyers on issues related to countering cybercrime in different countries of the world.

4. Results

Currently, cyberattacks are one of the most indicative ways of expressing cybercrime, which is characterized as a type of hostile action in which data or information systems are stolen, altered or destroyed in various ways. Such illegal activities are usually aimed at disrupting the operation of computers, information systems, information infrastructure, computer networks or a specific personal computer. Figure 1 shows the dynamics of the number of massive cyberattacks that took place in different countries in 2019-2020.

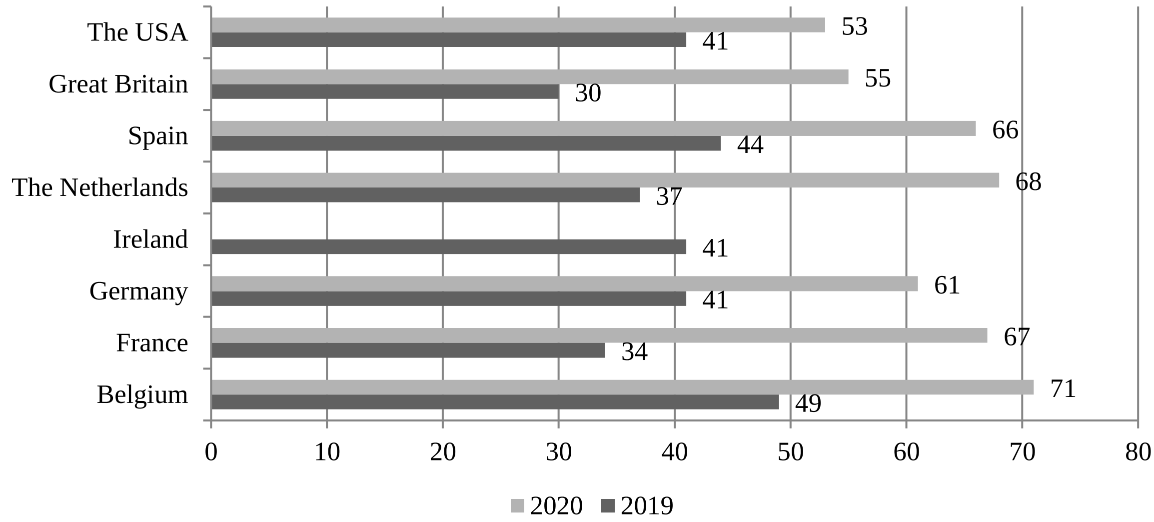

Figure 1 The number of massive cyberattacks against information and computer systems of companies around the world in 2019-2020, pcs. Source: (Perwej et al., 2021).

Organizations are increasingly in need of professional assistance on the part of researchers and cybersecurity practitioners to combat the numerous manifestations of cybercrime, which, having different technical characteristics, are constantly changing, improving, and causing increasing harm to business structures. This is due to the significant number and positive growing dynamics of cyberattacks in the world. Over 12 months of 2020, the average cost of preventing and responding to cybercrime and software breaches in the United States increased to 57,000 USD. This exceeds the amount of 10,000 USD reported in 2019 by almost six times. Of all the kinds of this type of crime, criminals mostly use phishing and malware infection nowadays. Large companies pay the highest prices for their presence and secure operation on the Internet since they are the most desirable targets for cyberattacks, and, therefore, cybercrime directed at them is much more expensive and intense (Perwej et al., 2021; Hajj et al., 2021).

Companies around the world suffer from numerous forms of cybersecurity breaches. Minor incidents and violations of official norms of work in cyberspace also create many problems for organizations (Figure 2).

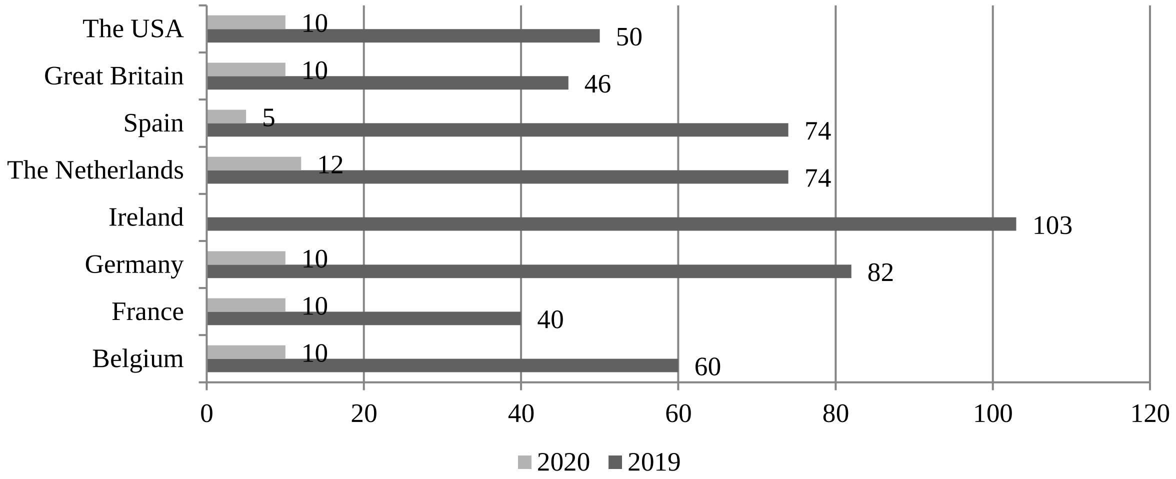

Figure 2 Minor incidents and breaches of cybersecurity at enterprises and organizations around the world in 2019-2020.
Source: (Perwej et al., 2021).

In addition to the increase in financial expenses of companies aimed at combating cybersecurity breaches, the dynamics of the growing number of cybercrimes in the world lead to an increase in the number and content of challenges related to the need for legal regulation in this area. Legal regulation of the fight against cybercrime in the world can be divided into several main periods (Figure 3):

1. The period of the emergence of legislation on combating cybercrime (1986-1989) – from the moment of adoption of the first-ever law on computer systems, which was crucial for the further development of legislation on combating cybercrime and became the impetus for the development of administrative legislation in European countries.
2. The period of active changes in the criminal and administrative law of European countries (1989-2000). After 1989, the rapid development of criminal and administrative legislation in European countries was initiated to strengthen the fight against computer crime, which continues to some extent to this day. This period is limited to 2000, after which amendments to national legislation were no longer widespread.
3. The stage of consolidation of the European community in combating cybercrime (2000-2001) – within two years, regulatory acts were introduced that significantly affected the quality and effectiveness of such a fight.
4. The current stage of legal support for combating cybercrime (2001 – present time) is characterized by the process of improving the legislative framework for regulating cybercrime in less developed countries and its progressive legal changes at the level of states and interstate structures and associations.

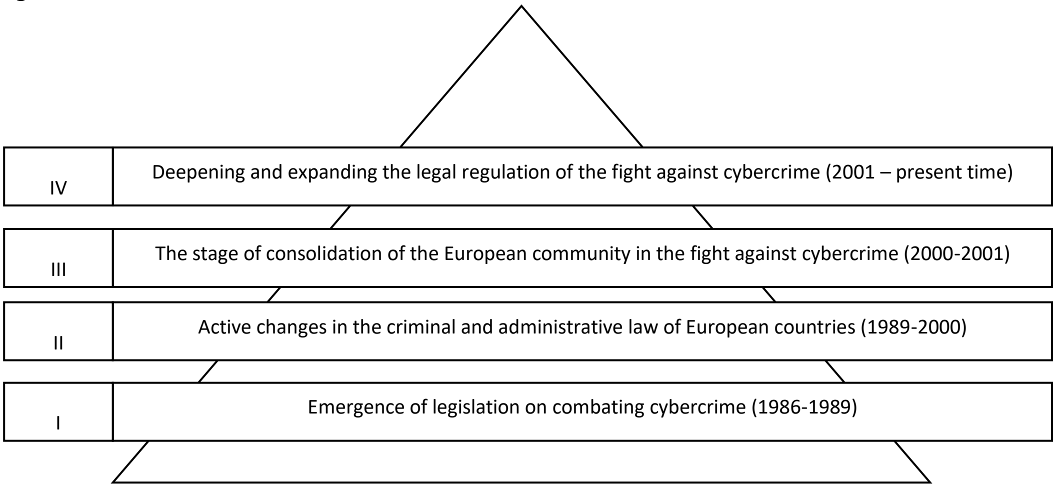

Figure 3 Stages of developing legal regulation in the fight against cybercrime in the world
Source: (Moreno et al., 2018; Perwej et al., 2021).

Taking into account the unlimited possibilities of the Internet and, in the vast majority of cases, the investigation of cybercrime by several states simultaneously, international European organizations are gaining special importance as subjects of legal regulation in the fight against cybercrime nowadays. In particular, the researchers point to the role of the Computer Emergency Response Team (CERT) in Brussels and the European Cybercrime Center in the Hague, as well as ENISA (Viraja and Purandare, 2021).

The mission of such organizations is primarily to combat new viruses in the global network and identify vulnerabilities in information security systems, develop strategic directions for combating them that can be implemented at the European level, and conduct cybersecurity seminars in cooperation with business, civil society and government agencies in the EU. The range of activities of these bodies includes active work in the theoretical, diplomatic and practical dimensions, providing technical support to investigations at the international and regional levels (Lallie et al., 2021; Cremer et al., 2022).

In recent years, the legal aspect of combating cybercrime in the EU can be characterized by the following features:

1) development and permanent improvement of legislation on combating cybercrime;
2) activities to combat cybercrime are carried out by both national and international organizations, which include the best specialists of the countries participating in such cooperation;
3) the study of theoretical concerns, such as expert evaluation of cybercrime and the introduction of new software and hardware instruments to reveal this sort of crime, plays a significant role in the fight against cybercrime;
4) the implementation of active information exchange is becoming more systematic and covers more and more countries.

When evaluating the most common types of cybercrime nowadays, it is worth noting that the Council of Europe Convention on Cybercrime identifies four main types of this type of legal offence:

- offences involving illegal access, illegal interception, data intrusion, system intrusion, and illegal use of equipment;
- computer crimes – computer forgery, computer fraud;
- content crimes – crimes related to pornography, personal information;
- offences related to infringement of copyright and related rights (Hazizova, 2023; Samunnisa et al., 2023).

Legislation in most countries of the world defines the following threats to the information sphere:

- violation of a person's right to access information resources;
- dissemination of information containing violence and cruelty;
- disclosure of information containing elements of state secrets, as well as various types of confidential data;
- attempts to manipulate public opinion, including by disseminating false, incomplete or biased information.

The information and analytical activities of government agencies, business structures and special companies, as well as specialists implementing the policy of combating cybercrime, include as follows:

- identifying the causes and factors that contribute to the spread of crime in cyberspace, as well as identifying the perpetrators of crimes;
- obtaining data on trends in the development of the criminal environment, concentrating this data and implementing administrative measures against offenders;
- organizing a system of information exchange with other divisions of the company, law enforcement agencies and control bodies, implementing measures to provide reasonable and reliable information to prevent crime, identify and prosecute perpetrators of criminal offences, and neutralize organized crime;
- identifying and evaluating problems that arise in the course of practical activities on the detection and prevention of offences;
- development of optimal management solutions for specific problems and measures to address them (Saleous et al., 2023).

Significant aspects of the Council of Europe Convention on Cybercrime are the definition and updating of the following directions of European cybersecurity policy:

1) educational direction, which aims to raise consumers' awareness regarding cyber threats;
2) creation of a full-fledged system of warning and informing about new threats in cyberspace;
3) technical, information and technological support for combating cybercrime;
4) standardization and certification of cybercrime and cybersecurity tools;
5) legal support in combating crimes in cyberspace;
6) strengthening security at the state level;
7) International cooperation on information security issues (Tymchyshyn, 2020).

In addition, the tasks of ensuring information security in cyberspace include as following:

- promoting the introduction of state-of-the-art information technologies and the production of a competitive national information product, including modern means and systems for protecting information resources;
- ensuring the security of information and telecommunication systems in the interests of public administration, meeting the needs of defence and security of the state, credit, banking and other sectors of the economy, as well as critical infrastructure management systems;
- creation of a national cyber security system (Sarre et al., 2018; Lelyk, 2022).

It can be concluded from the aforesaid that one of the key vectors of the fight against cybercrime nowadays is the optimization of legal regulation of combating cybercrime with special attention being paid to the direction of legal support for this type of activity.

Assessing the scientists' standpoint on the conditions for effective counteraction to cybercrime makes it possible to identify the prerequisites and factors that pose threats to cyberspace security at this stage of developing legal regulation of this sphere in most countries of the world, namely:

- imperfection of the regulatory framework in the field of cybersecurity and information protection, as well as its obsolescence;
- slow implementation of the provisions of European law in the legislation of particular countries, insufficient regulation of the digital component of crime investigation, as well as low level of legal liability for violation of the law in this sphere;
- a significant number of ministries and agencies do not have relevant structural units, the necessary staffing and proper control over the cyber defence while funding for cyber defence is allocated on a residual basis;
- the lack of an independent information security testing system and mechanisms for disclosing information about vulnerabilities in the context of the dynamic digitalization of all spheres of public administration and the country's life, which requires strict adherence to relevant standards;
- inconsistency with modern requirements to the level of education and training of specialists in cybersecurity and cyber defence, including ineffective mechanisms for encouraging them to work in the public sector;
- lack of legislation on critical infrastructure and its protection, which significantly complicates the formation of a cyber defence system for this infrastructure;
- the incompleteness of measures to implement an organizational and technical model of cyber defence that meets modern threats, challenges in cyberspace and global trends in developing the cybersecurity sector;
- the lack of a system to improve citizens' digital literacy and the culture of safe behaviour in cyberspace, which contributes to raising public awareness of cyber threats and cyber defence (Vasylyev et al., 2021).

In view of the foregoing, it should be emphasized that the major challenges for many countries in the world in the field of cybersecurity nowadays are the active use of cyber tools in international competition for world leadership; the competitive nature of the development of cybersecurity tools and the implementation of cyber threats in the process of rapidly progressing information changes in communication technologies, cloud computing, and database formation; growth of technological capabilities of cyber weapons, which enables the enemy to covertly carry out cyber attacks and cyber operations, remotely take control of control systems, damage and destroy critical information infrastructure; increasing the technological level of unlawful interference through the use of social engineering methods, artificial intelligence and crypto technologies; the impact of the spread of the COVID-19 pandemic and hostilities in the world on economic activity and social behavior of criminals, which, due to the widespread use of electronic services, led to the rapid transformation and organization of a significant part of social relations (Sarre et al., 2018).

The strategy for developing a country's cybersecurity usually stipulates that the main areas of security in cyberspace are as follows:

- development of a safe, stable and reliable cyberspace, that is, the creation of a unified regulatory framework and its deployment among the general public for increasing public awareness;
- cybersecurity of state electronic information resources and information processing infrastructure, the need for protection of which is determined by law;
- development of effective countermeasures at the state and local government level;
- cyber defence of critical infrastructures, which provides for the development of a unified mechanism of public-private partnership to prevent cyber threats;
- development of the capacity of the security and defence sector in all previous categories and their integration into a unified complex (Kuzior et al., 2022).

MONEYVAL's typological study "Criminal money flows on the Internet: methods, trends and multi-stakeholder counteraction" examined the following risks of cybercrime and money laundering:

- technical;
- operational;
- legal;
- geographical;
- judicial (Danich, 2019).

At the same time, such classifications are slightly generalized and require more detailed consideration, taking into account the nature of threats and vulnerability of society and the state to cybercrime, as well as the increasing negative consequences of such crimes, the need to find ways to counteract such offences or reduce their impact.

5. Discussion

Currently, cybersecurity is recognized as a political, economic, and social threat that is constantly evolving around the world. The most frequent crimes in cyberspace are disruption of computer systems of business and government organizations and hacking of databases. Other common types of cybercrime are theft of technology or innovation and, obviously, theft of money.

It is worth recognizing that the rapid development in the technological sphere gives rise to new types of services, including financial services. One of the most prevalent types of fraud involves thieves stealing data regarding wage payments from businesses and selling it on the black market. This, in turn, forces offenders to look for new ways to make a profit in the cyber environment (Slobodianyk, 2022).

Based on characterizing the essence and types of cybercrime most common in the modern information space, the following most significant aspects can be identified as prerequisites for the threats of cybercrime spreading both in society as a whole and in the work of state authorities:

- openness of the society and state. The user-friendly infrastructure created based on computer networks and information technologies for the international delivery of goods, provision of services, money transfers between individuals and legal entities, storage of information on the Internet and connection of any computer to it, at the same time, provides various opportunities for committing cybercrime;
- high speed and low cost of crime. The aforementioned infrastructure provides criminals with extremely fast access to any information, documents and ultimately private property while providing inexpensive, efficient and virtually anonymous payment options;
- high level of technological efficiency. The rapid progress of the information environment, as well as a relatively long bureaucratic approach to the establishment of the regulatory framework, led to a significant lag between measures to prevent and combat cybercrime and the pace of development of criminal acts;
- Complexity of the nature of the offence. In addition to obtaining financial or other material benefits from committing a crime, cybercriminals use computer technologies, information and communication networks for social and psychological reasons, in particular, to discredit governments and states, to host terrorist websites, to damage key information systems by entering fake data to damage, destroy or completely disable these systems;
- the anonymity of the crime. Criminals are attracted by the lack of physical contact with the victim, the relatively lenient penalties in some countries, and, of course, the difficulty of recognizing, capturing and seizing forensically relevant information in the virtual space;
- transnational and massive nature of crime. The feature of this type of crime is that it can be organized and committed almost without any space restrictions. Moreover, cybercrime is becoming more popular since computers and Internet services become more accessible to increasing numbers of people (Danich, 2019).

Priority and most crucial strategic actions to prevent cybercrime include developing the capacity of criminal justice and law enforcement agencies, adopting strategies and laws to combat cybercrime, and creating a strong knowledge base and cooperation between government administrative structures. At the same time, effective state governance of security issues, information and educational activities, and intensification of security activities of the public community at the private, state and international levels should be ensured.

Even if the country's leadership is strategically focused on combating cybercrime, in many cases the state does not have the necessary technical capabilities to develop the relevant laws or mechanisms for their implementation when such laws already exist. Criminals can anonymously conduct crimes through a network in an underdeveloped country, using unauthorized access, etc., and crimes committed abroad remain unpunished because of the absence of the necessary legislative framework and technological capabilities for its implementation (Bhati et al., 2020).

This is precisely why creating a unified strategy for information exchange and improving the ability to detect cybercriminals is more important than ever (Elmasry et al., 2019).

There are several other regulatory issues related to countering cyber threats, which require the state to go beyond a purely state-centric approach and apply new methods that focus on effective public-private partnerships to organize an effective fight against cyber espionage. For instance, law enforcement agencies cannot effectively combat cybercrime if similar functions and responsibilities are not concentrated in their hands but are distributed among several ministries and agencies. However, official public-private partnership networks created in this context can be an effective type of interaction in this area. Formality and regulation of such cooperation are of particular importance since both parties to such partnerships are not inclined to share important information, especially when we are talking about international and foreign private companies. Such cooperation should involve not only private companies operating in the so-called "critical" areas of activity, but also organizations specializing in information security, as well as software developers, device manufacturers, and electronic payment service providers (Danich, 2019).

It is worth noting that the activities of cybercriminals are punishable under articles of administrative and criminal law. Currently, it is necessary to develop a common strategy for cooperation in information exchange and improving the ability to search for cybercriminals. There are several other issues of democratic governance related to countering cyber threats (Moro, 2020; Danich, 2019).

6. Conclusions

Thus, the conducted analysis of the scientific literature on the research topic has shown that combating cybercrime is extremely relevant nowadays. The academic paper provides evidence of a steady upward trend in the number of cybercrimes in many countries of the world. Such a situation shows that increasing the level of information security requires strengthening both strategic planning of measures to counter cybercrime at the national and international levels and optimization of operational response to counter cybercrime to optimize the functioning of the economic and social system and its development. The current state of global security encourages the development of a new state model of cybersecurity both in individual countries and on a global scale, where any type of activity cannot go unprotected in an environment where cyber threats are constantly emerging and actively developing.

Ethical considerations

Not applicable.

Conflict of Interest

The authors declare no conflicts of interest.

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